Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Kronberg Facility

Physical address of dairy:

20777 Riverside AVE	Hilmar	Stanislaus	95324
Number and Street	City	County	Zip Code

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1950

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X045-X262-X001-XXXX X045-X360-X022-XXXX

B. OPERATORS

Kronberg, Virgil

Operator name: Kronberg, Virgil

Telephone no.: Landline: ; Cellular: (209) 620-4520

20931 Riverside AVE	Hilmar	CA	95324
Mailing Address Number and Street	City	State	Zip Code

This operator is responsible for paying permit fees.

C. OWNERS

Kronberg, Virgil

Legal owner name: Kronberg, Virgil

Telephone no.: Landline: ; Cellular: (209) 620-4520

20931 Riverside AVE	Hilmar	CA	95324
Mailing Address Number and Street	City	State	Zip Code

This owner is responsible for paying permit fees.

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	0	0	0	0
Number under roof	0	0	0	0	0	0
Maximum number	0	0	0	0	0	0
Average number	0	0	0	0	0	0
Avg live weight (lbs)	0	0	0	0		

Predominant milk cow breed: Holstein

Average milk production: 1 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 1.00 *tons per reporting period*

Total nitrogen from manure: 1.00 *lbs per reporting period*

After ammonia losses (30% loss applied): 0.70 *lbs per reporting period*

Total phosphorus from manure: 1.00 *lbs per reporting period*

Total potassium from manure: 1.00 *lbs per reporting period*

Total salt from manure: 0.00 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: -3 *gallons*

Total nitrogen generated: -0.02 *lbs*

Total phosphorus generated: 0.00 *lbs*

Total potassium generated: -0.02 *lbs*

Total salt generated: -0.12 *lbs*

	9,109,997	*gallons applied*
+	0	*gallons exported*
-	9,110,000	*gallons imported*
=	-3	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Domestic Well - Dairy Well	Ground water
Irrigation water - canal	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

Date	Material type / Description	Quantity	N (mg/L)	P (mg/L)	K (mg/L)	Salt (mg/L)	TDS (mg/L)
10/15/2023	Process wastewater *LWI - WJF*	9,110,000.00 *gal*	641.00	84.00	741.00		4,880

No commercial or other nutrient imports entered.

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	48,730.71	6,385.93	56,333.01	370,992.00
Total imports for all materials	48,730.71	6,385.93	56,333.01	370,992.00

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
1	14	14	2	process wastewater	X045-X360-X022-XXXX
2	30	30	2	process wastewater	X045-X360-X022-XXXX
3	11	11	2	process wastewater	X045-X360-X022-XXXX
4	19	19	2	process wastewater	X045-X262-X001-XXXX
Totals for areas that were used for application	74	74	8		
Totals for areas that were not used for application					
Land application area totals	74	74	8		

B. CROPS AND HARVESTS

1

Field name: 1

11/03/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 14 Plant date: 11/03/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/16/2023	175.25 *ton*	Dry-weight		62.6	27,300.00	4,000.00	31,300.00		10.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	12.52	255.62	37.45	293.07	1,020.61

06/13/2023: Corn, silage

Crop: Corn, silage Acres planted: 14 Plant date: 06/13/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/04/2023	504.00 *ton*	Dry-weight		68.4	13,000.00	2,400.00	15,700.00		5.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	36.00	295.78	54.60	357.21	1,183.10

2

Field name: 2

11/03/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 30 Plant date: 11/03/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/16/2023	600.00 *ton*	Dry-weight		62.2	12,700.00	1,900.00	14,700.00		7.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	20.00	192.02	28.73	222.26	1,134.00

06/13/2023: Corn, silage

Crop: Corn, silage Acres planted: 30 Plant date: 06/13/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/04/2023	1,080.00 *ton*	Dry-weight		65.6	13,700.00	2,000.00	13,100.00		5.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	36.00	339.32	49.54	324.46	1,312.70

3

Field name: 3

11/03/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 11 Plant date: 11/03/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/16/2023	150.00 *ton*	Dry-weight		62.6	27,300.00	4,000.00	31,300.00		10.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	13.64	278.46	40.80	319.26	1,111.80

3

06/13/2023: Corn, silage

Crop: Corn, silage Acres planted: 11 Plant date: 06/13/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/04/2023	396.00 *ton*	Dry-weight		68.4	13,000.00	2,400.00	15,700.00		5.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	36.00	295.78	54.60	357.21	1,183.10

4

Field name: 4

11/03/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 19 Plant date: 11/03/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/16/2023	380.00 *ton*	Dry-weight		63.5	19,300.00	3,100.00	24,500.00		10.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	20.00	281.78	45.26	357.70	1,460.00

06/13/2023: Corn, silage

Crop: Corn, silage Acres planted: 19 Plant date: 06/13/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/04/2023	658.50 *ton*	Dry-weight		69.6	13,200.00	2,800.00	13,000.00		4.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	34.66	278.15	59.00	273.94	1,032.53

NUTRIENT BUDGET

A. LAND APPLICATIONS

1 - 11/03/2022: Oats, silage-soft dough

Field name: 1

Crop: Oats, silage-soft dough Plant date: 11/03/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	104.09	13.64	120.33	792.46	272,432.00 *gal*
Irrigation water - canal	Surface water	0.00	0.00	0.00	38.84	1,515,375.00 *gal*
Application event totals		104.09	13.64	120.33	831.30	

04/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	95.42	12.50	110.30	726.42	249,730.00 *gal*
Irrigation water - canal	Surface water	0.00	0.00	0.00	38.84	1,515,375.00 *gal*
Application event totals		95.42	12.50	110.30	765.26	

10/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	112.77	14.78	130.36	858.50	295,135.00 *gal*
Irrigation water - canal	Surface water	0.00	0.00	0.00	41.43	1,616,400.00 *gal*
Application event totals		112.77	14.78	130.36	899.93	

1 - 06/13/2023: Corn, silage

Field name: 1

Crop: Corn, silage Plant date: 06/13/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

1 - 06/13/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation water - canal	Surface water	0.00	0.00	0.00	41.43	1,616,400.00 *gal*
Application event totals		0.00	0.00	0.00	41.43	

07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	86.74	11.37	100.28	660.38	227,027.00 *gal*
Irrigation water - canal	Surface water	0.00	0.00	0.00	38.84	1,515,375.00 *gal*
Application event totals		86.74	11.37	100.28	699.22	

07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	86.74	11.37	100.28	660.38	227,027.00 *gal*
Irrigation water - canal	Surface water	0.00	0.00	0.00	38.84	1,515,375.00 *gal*
Application event totals		86.74	11.37	100.28	699.22	

08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	86.02	11.27	99.44	654.88	225,135.00 *gal*
Irrigation water - canal	Surface water	0.00	0.00	0.00	38.84	1,515,375.00 *gal*
Application event totals		86.02	11.27	99.44	693.72	

08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	86.74	11.37	100.28	660.38	227,027.00 *gal*
Irrigation water - canal	Surface water	0.00	0.00	0.00	38.84	1,515,375.00 *gal*
Application event totals		86.74	11.37	100.28	699.22	

1 - 06/13/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation water - canal	Surface water	0.00	0.00	0.00	38.84	1,515,375.00 *gal*
Application event totals		0.00	0.00	0.00	38.84	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation water - canal	Surface water	0.00	0.00	0.00	41.43	1,616,400.00 *gal*
Application event totals		0.00	0.00	0.00	41.43	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation water - canal	Surface water	0.00	0.00	0.00	41.43	1,616,400.00 *gal*
Application event totals		0.00	0.00	0.00	41.43	

2 - 11/03/2022: Oats, silage-soft dough

Field name: 2

Crop: Oats, silage-soft dough

Plant date: 11/03/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	52.05	6.82	60.17	396.23	291,892.00 *gal*
Irrigation water - canal	Surface water	0.00	0.00	0.00	33.83	2,828,700.00 *gal*
Application event totals		52.05	6.82	60.17	430.06	

2 - 11/03/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	95.42	12.50	110.30	726.42	535,135.00 *gal*
Irrigation water - canal	Surface water	0.00	0.00	0.00	33.83	2,828,700.00 *gal*
Application event totals		95.42	12.50	110.30	760.26	

10/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	112.77	14.78	130.36	858.50	632,432.00 *gal*
Irrigation water - canal	Surface water	0.00	0.00	0.00	30.21	2,525,625.00 *gal*
Application event totals		112.77	14.78	130.36	888.71	

2 - 06/13/2023: Corn, silage

Field name: 2

Crop: Corn, silage

Plant date: 06/13/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation water - canal	Surface water	0.00	0.00	0.00	33.83	2,828,700.00 *gal*
Application event totals		0.00	0.00	0.00	33.83	

07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	138.79	18.19	160.44	1,056.61	778,378.00 *gal*
Irrigation water - canal	Surface water	0.00	0.00	0.00	30.21	2,525,625.00 *gal*
Application event totals		138.79	18.19	160.44	1,086.82	

2 - 06/13/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	86.74	11.37	100.28	660.38	486,486.00 *gal*
Irrigation water - canal	Surface water	0.00	0.00	0.00	30.21	2,525,625.00 *gal*
Application event totals		86.74	11.37	100.28	690.59	

08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	86.02	11.27	99.44	654.88	482,432.00 *gal*
Irrigation water - canal	Surface water	0.00	0.00	0.00	30.21	2,525,625.00 *gal*
Application event totals		86.02	11.27	99.44	685.09	

08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	86.74	11.37	100.28	660.38	486,486.00 *gal*
Irrigation water - canal	Surface water	0.00	0.00	0.00	30.21	2,525,625.00 *gal*
Application event totals		86.74	11.37	100.28	690.59	

08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation water - canal	Surface water	0.00	0.00	0.00	30.21	2,525,625.00 *gal*
Application event totals		0.00	0.00	0.00	30.21	

09/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation water - canal	Surface water	0.00	0.00	0.00	30.21	2,525,625.00 *gal*
Application event totals		0.00	0.00	0.00	30.21	

2 - 06/13/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation water - canal	Surface water	0.00	0.00	0.00	30.21	2,525,625.00 *gal*
Application event totals		0.00	0.00	0.00	30.21	

3 - 11/03/2022: Oats, silage-soft dough

Field name: 3

Crop: Oats, silage-soft dough

Plant date: 11/03/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	104.09	13.64	120.33	792.46	214,054.00 *gal*
Irrigation water - canal	Surface water	0.00	0.00	0.00	65.91	2,020,500.00 *gal*
Application event totals		104.09	13.64	120.33	858.37	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	95.42	12.50	110.30	726.42	196,216.00 *gal*
Irrigation water - canal	Surface water	0.00	0.00	0.00	65.91	2,020,500.00 *gal*
Application event totals		95.42	12.50	110.30	792.33	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	112.77	14.78	130.36	858.50	231,892.00 *gal*
Irrigation water - canal	Surface water	0.00	0.00	0.00	72.50	2,222,550.00 *gal*
Application event totals		112.77	14.78	130.36	931.00	

3 - 06/13/2023: Corn, silage

3 - 06/13/2023: Corn, silage

Field name: 3

Crop: Corn, silage

Plant date: 06/13/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation water - canal	Surface water	0.00	0.00	0.00	65.91	2,020,500.00 *gal*
Application event totals		0.00	0.00	0.00	65.91	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	86.74	11.37	100.28	660.38	178,378.00 *gal*
Irrigation water - canal	Surface water	0.00	0.00	0.00	72.50	2,222,550.00 *gal*
Application event totals		86.74	11.37	100.28	732.88	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	86.74	11.37	100.28	660.38	178,378.00 *gal*
Irrigation water - canal	Surface water	0.00	0.00	0.00	72.50	2,222,550.00 *gal*
Application event totals		86.74	11.37	100.28	732.88	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	86.02	11.27	99.44	654.88	176,892.00 *gal*
Irrigation water - canal	Surface water	0.00	0.00	0.00	72.50	2,222,550.00 *gal*
Application event totals		86.02	11.27	99.44	727.38	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	86.74	11.37	100.28	660.38	178,378.00 *gal*
Irrigation water - canal	Surface water	0.00	0.00	0.00	72.50	2,222,550.00 *gal*
Application event totals		86.74	11.37	100.28	732.88	

3 - 06/13/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation water - canal	Surface water	0.00	0.00	0.00	72.50	2,222,550.00 *gal*
Application event totals		0.00	0.00	0.00	72.50	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation water - canal	Surface water	0.00	0.00	0.00	72.50	2,222,550.00 *gal*
Application event totals		0.00	0.00	0.00	72.50	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation water - canal	Surface water	0.00	0.00	0.00	72.50	2,222,550.00 *gal*
Application event totals		0.00	0.00	0.00	72.50	

4 - 11/03/2022: Oats, silage-soft dough

Field name: 4

Crop: Oats, silage-soft dough

Plant date: 11/03/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	104.09	13.64	120.33	792.46	369,730.00 *gal*
Irrigation water - canal	Surface water	0.00	0.00	0.00	38.16	2,020,500.00 *gal*
Application event totals		104.09	13.64	120.33	830.62	

4 - 11/03/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	95.42	12.50	110.30	726.42	338,919.00 *gal*
Irrigation water - canal	Surface water	0.00	0.00	0.00	38.16	2,020,500.00 *gal*
Application event totals		95.42	12.50	110.30	764.58	

10/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	112.77	14.78	130.36	858.50	400,541.00 *gal*
Irrigation water - canal	Surface water	0.00	0.00	0.00	36.25	1,919,475.00 *gal*
Application event totals		112.77	14.78	130.36	894.75	

4 - 06/13/2023: Corn, silage

Field name: 4

Crop: Corn, silage

Plant date: 06/13/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation water - canal	Surface water	0.00	0.00	0.00	38.16	2,020,500.00 *gal*
Application event totals		0.00	0.00	0.00	38.16	

07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	86.74	11.37	100.28	660.38	308,108.00 *gal*
Irrigation water - canal	Surface water	0.00	0.00	0.00	36.25	1,919,475.00 *gal*
Application event totals		86.74	11.37	100.28	696.63	

4 - 06/13/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	86.74	11.37	100.28	660.38	308,108.00 *gal*
Irrigation water - canal	Surface water	0.00	0.00	0.00	36.25	1,919,475.00 *gal*
Application event totals		86.74	11.37	100.28	696.63	

08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	86.02	11.27	99.44	654.88	305,541.00 *gal*
Irrigation water - canal	Surface water	0.00	0.00	0.00	36.25	1,919,475.00 *gal*
Application event totals		86.02	11.27	99.44	691.13	

08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	86.74	11.37	100.28	660.38	308,108.00 *gal*
Irrigation water - canal	Surface water	0.00	0.00	0.00	36.25	1,919,475.00 *gal*
Application event totals		86.74	11.37	100.28	696.63	

08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation water - canal	Surface water	0.00	0.00	0.00	36.25	1,919,475.00 *gal*
Application event totals		0.00	0.00	0.00	36.25	

09/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation water - canal	Surface water	0.00	0.00	0.00	36.25	1,919,475.00 *gal*
Application event totals		0.00	0.00	0.00	36.25	

4 - 06/13/2023: Corn, silage

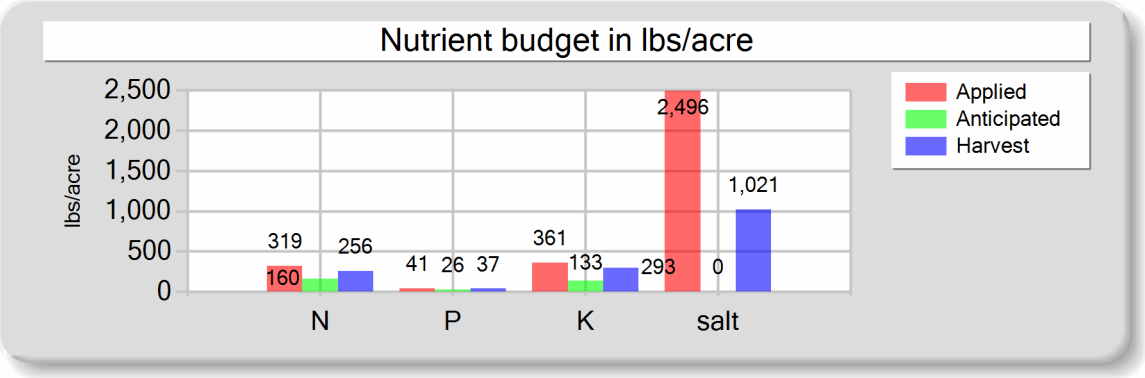
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation water - canal	Surface water	0.00	0.00	0.00	36.25	1,919,475.00 *gal*
Application event totals		0.00	0.00	0.00	36.25	

B. NUTRIENT BUDGET

1 - 11/03/2022: Oats, silage-soft dough

Field name: 1 Crop: Oats, silage-soft dough Plant date: 11/03/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	312.27	40.92	360.99	2,377.38
Fresh water	0.00	0.00	0.00	119.11
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	319.27	40.92	360.99	2,496.49
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	255.62	37.45	293.07	1,020.61
Nutrient balance	63.65	3.47	67.92	1,475.88
Applied to removed ratio	1.25	1.09	1.23	2.45

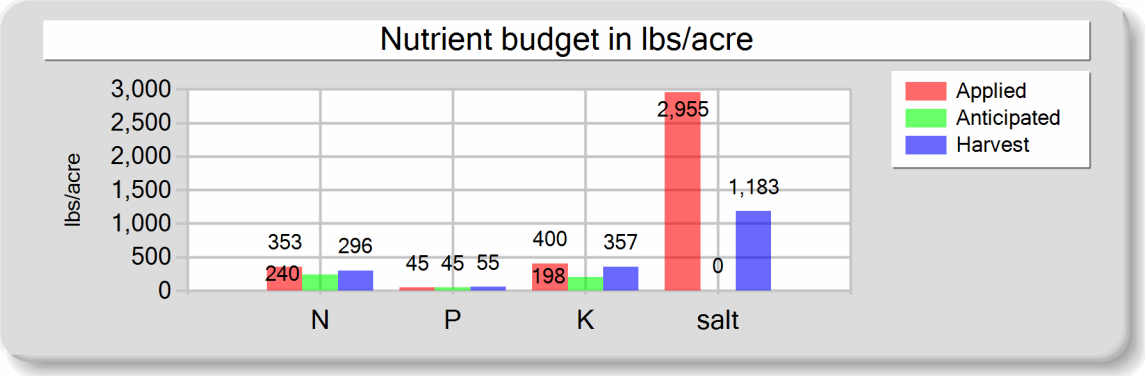
Fresh water applied	
4,647,150.00	*gallons*
171.14	*acre-inches*
12.22	*inches/acre*

Process wastewater applied	
817,297.00	*gallons*
30.10	*acre-inches*
2.15	*inches/acre*

Total harvests for the crop	
1	*harvests*

1 - 06/13/2023: Corn, silage

Field name: 1 Crop: Corn, silage Plant date: 06/13/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	346.25	45.37	400.27	2,636.03
Fresh water	0.00	0.00	0.00	318.49
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	353.25	45.37	400.27	2,954.52
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	295.78	54.60	357.21	1,183.10
Nutrient balance	57.47	-9.23	43.06	1,771.42
Applied to removed ratio	1.19	0.83	1.12	2.50

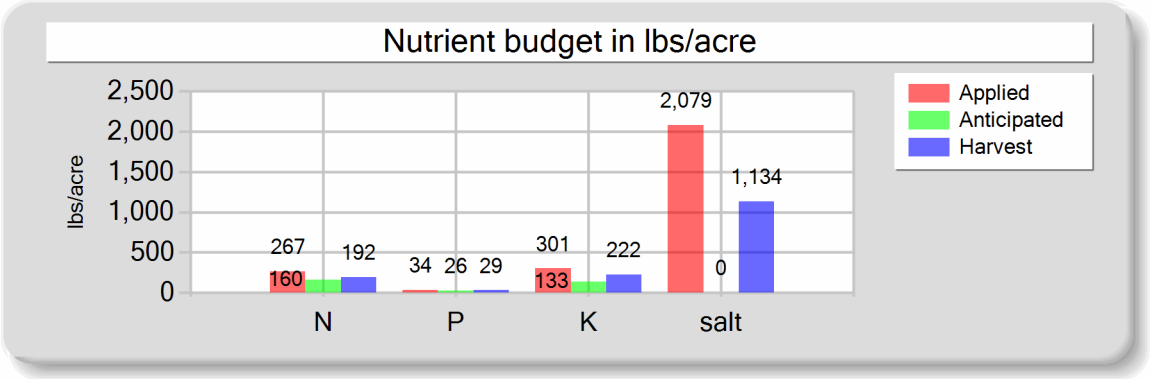
Fresh water applied	
12,426,075.00	*gallons*
457.61	*acre-inches*
32.69	*inches/acre*

Process wastewater applied	
906,216.00	*gallons*
33.37	*acre-inches*
2.38	*inches/acre*

Total harvests for the crop	
1	*harvests*

2 - 11/03/2022: Oats, silage-soft dough

Field name: 2 Crop: Oats, silage-soft dough Plant date: 11/03/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	260.23	34.10	300.83	1,981.15
Fresh water	0.00	0.00	0.00	97.88
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	267.23	34.10	300.83	2,079.03
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	192.02	28.73	222.26	1,134.00
Nutrient balance	75.20	5.37	78.56	945.03
Applied to removed ratio	1.39	1.19	1.35	1.83

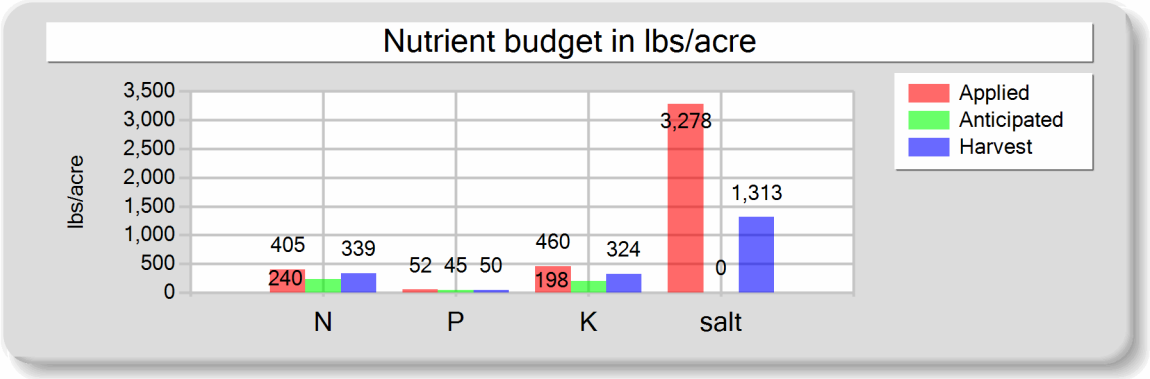
Fresh water applied	
8,183,025.00	*gallons*
301.35	*acre-inches*
10.05	*inches/acre*

Process wastewater applied	
1,459,459.00	*gallons*
53.75	*acre-inches*
1.79	*inches/acre*

Total harvests for the crop	
1	*harvests*

2 - 06/13/2023: Corn, silage

Field name: 2 Crop: Corn, silage Plant date: 06/13/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	398.29	52.19	460.43	3,032.25
Fresh water	0.00	0.00	0.00	245.30
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	405.29	52.19	460.43	3,277.56
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	339.32	49.54	324.46	1,312.70
Nutrient balance	65.97	2.66	135.97	1,964.85
Applied to removed ratio	1.19	1.05	1.42	2.50

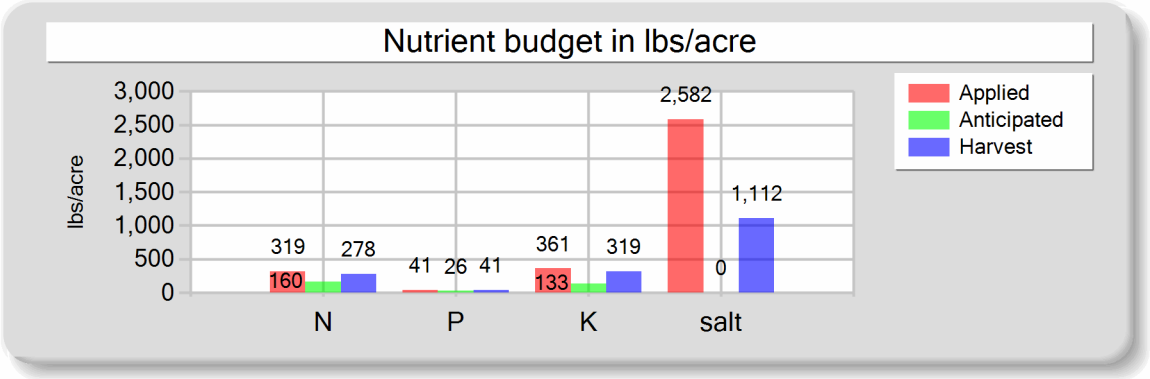
Fresh water applied	
20,508,075.00	*gallons*
755.24	*acre-inches*
25.17	*inches/acre*

Process wastewater applied	
2,233,782.00	*gallons*
82.26	*acre-inches*
2.74	*inches/acre*

Total harvests for the crop	
1	*harvests*

3 - 11/03/2022: Oats, silage-soft dough

Field name: 3 Crop: Oats, silage-soft dough Plant date: 11/03/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	312.27	40.92	360.99	2,377.38
Fresh water	0.00	0.00	0.00	204.33
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	319.27	40.92	360.99	2,581.70
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	278.46	40.80	319.26	1,111.80
Nutrient balance	40.81	0.12	41.73	1,469.90
Applied to removed ratio	1.15	1.00	1.13	2.32

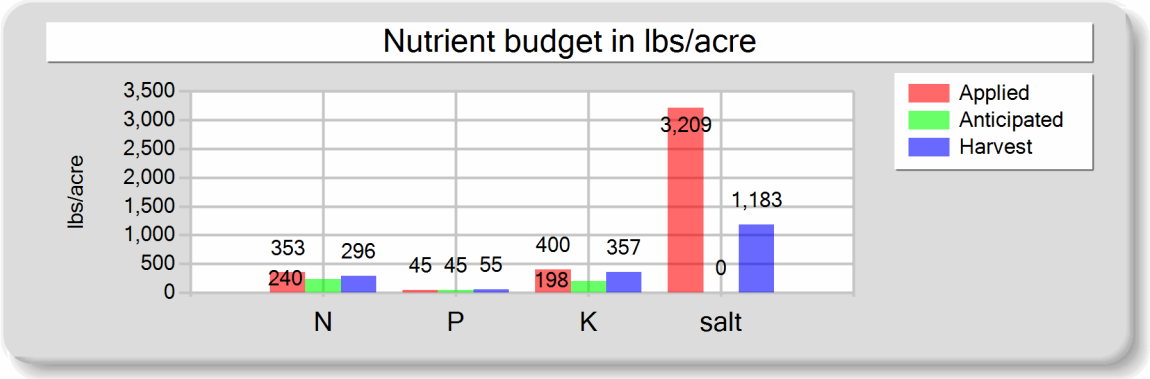
Fresh water applied	
6,263,550.00	*gallons*
230.67	*acre-inches*
20.97	*inches/acre*

Process wastewater applied	
642,162.00	*gallons*
23.65	*acre-inches*
2.15	*inches/acre*

Total harvests for the crop	
1	*harvests*

3 - 06/13/2023: Corn, silage

Field name: 3 Crop: Corn, silage Plant date: 06/13/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	346.25	45.37	400.27	2,636.02
Fresh water	0.00	0.00	0.00	573.43
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	353.25	45.37	400.27	3,209.45
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	295.78	54.60	357.21	1,183.10
Nutrient balance	57.47	-9.23	43.06	2,026.35
Applied to removed ratio	1.19	0.83	1.12	2.71

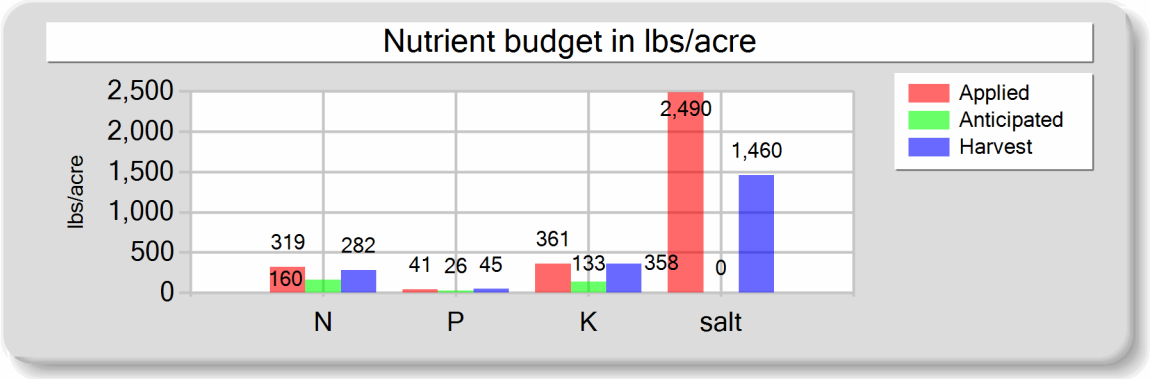
Fresh water applied	
17,578,350.00	*gallons*
647.35	*acre-inches*
58.85	*inches/acre*

Process wastewater applied	
712,026.00	*gallons*
26.22	*acre-inches*
2.38	*inches/acre*

Total harvests for the crop	
1	*harvests*

4 - 11/03/2022: Oats, silage-soft dough

Field name: 4 Crop: Oats, silage-soft dough Plant date: 11/03/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	312.27	40.92	360.99	2,377.38
Fresh water	0.00	0.00	0.00	112.57
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	319.27	40.92	360.99	2,489.95
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	281.78	45.26	357.70	1,460.00
Nutrient balance	37.49	-4.34	3.29	1,029.95
Applied to removed ratio	1.13	0.90	1.01	1.71

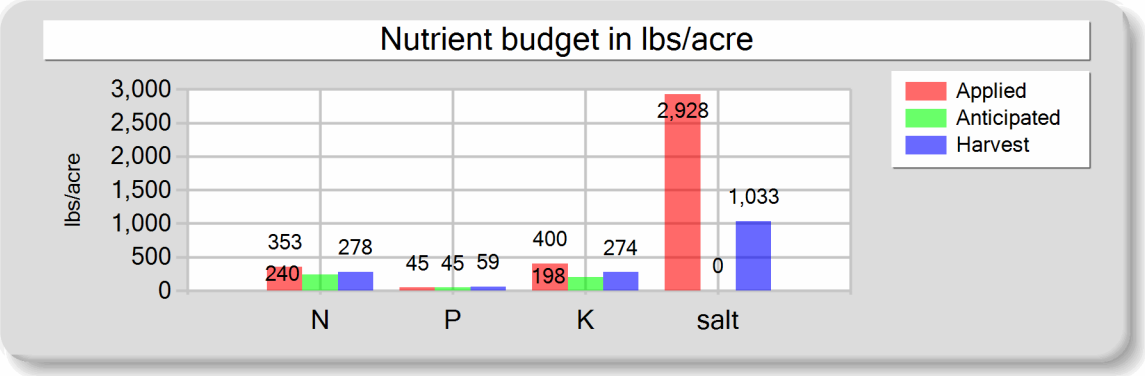
Fresh water applied	
5,960,475.00	*gallons*
219.50	*acre-inches*
11.55	*inches/acre*

Process wastewater applied	
1,109,190.00	*gallons*
40.85	*acre-inches*
2.15	*inches/acre*

Total harvests for the crop	
1	*harvests*

4 - 06/13/2023: Corn, silage

Field name: 4 Crop: Corn, silage Plant date: 06/13/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	346.25	45.37	400.27	2,636.03
Fresh water	0.00	0.00	0.00	291.92
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	353.25	45.37	400.27	2,927.95
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	278.15	59.00	273.94	1,032.53
Nutrient balance	75.10	-13.63	126.33	1,895.42
Applied to removed ratio	1.27	0.77	1.46	2.84

Fresh water applied	
15,456,825.00	*gallons*
569.22	*acre-inches*
29.96	*inches/acre*

Process wastewater applied	
1,229,865.00	*gallons*
45.29	*acre-inches*
2.38	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Solid Manure

Sample and source description: Solid Manure

Sample date: 05/18/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 9.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	2,600.00	1,400.00	1,900.00							
DL	100.00	100.00	100.00							

B. PROCESS WASTEWATER ANALYSES

LWI-WJF

Sample and source description: LWI-WJF

Sample date: 04/24/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	745.00	0.00	0.00	0.00	94.30	780.00								7,790.00	5,540
DL	0.10	0.10	0.10	0.10	0.10	0.10								10.00	10

LWI-WJF

Sample and source description: LWI-WJF

Sample date: 07/07/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	770.00	0.00	0.00	0.00	72.40	725.00								7,630.00	4,260
DL	0.10	0.10	0.10	0.10	0.10	0.10								10.00	10

LWI-WJF

Sample and source description: LWI-WJF

Sample date: 10/04/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	641.00	0.00	0.00	0.00	84.00	741.00								7,580.00	4,880
DL	0.10	0.10	0.10	0.10	0.10	0.10								10.00	10

C. FRESH WATER ANALYSES

Domestic Well - Dairy Well

Dairy Well

Sample description: Dairy Well

Sample date: 08/02/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value			23.00								1,140.00	
DL			1.00								10.00	

Irrigation water - canal

Irrigation water - canal water

Sample description: Irrigation water - canal water

Sample date: 08/02/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								30.50	43
DL	0.01	0.01	0.01								10.00	10

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

1 - 11/03/2022: Oats, silage-soft dough

1 + 3

Sample and source description: 1 + 3

Sample date: 05/19/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 62.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	27,300.00	4,000.00	31,300.00		10.90
DL	0.01	0.01	0.01		0.01

1 - 06/13/2023: Corn, silage

1 + 3

Sample and source description: 1 + 3

Sample date: 10/04/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,000.00	2,400.00	15,700.00		5.20
DL	100.00	100.00	100.00		0.01

2 - 11/03/2022: Oats, silage-soft dough

2

Sample and source description: 2

Sample date: 05/19/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 62.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,700.00	1,900.00	14,700.00		7.50
DL	0.01	0.01	0.01		0.01

2 - 06/13/2023: Corn, silage

2

Sample and source description: 2

Sample date: 10/04/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,700.00	2,000.00	13,100.00		5.30
DL	100.00	100.00	100.00		0.01

3 - 11/03/2022: Oats, silage-soft dough

1 + 3

Sample and source description: 1 + 3

Sample date: 05/19/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 62.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	27,300.00	4,000.00	31,300.00		10.90
DL	0.01	0.01	0.01		0.01

3 - 06/13/2023: Corn, silage

1 + 3

Sample and source description: 1 + 3

Sample date: 10/04/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,000.00	2,400.00	15,700.00		5.20
DL	100.00	100.00	100.00		0.01

4 - 11/03/2022: Oats, silage-soft dough

4

Sample and source description: 4

Sample date: 05/19/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 63.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,300.00	3,100.00	24,500.00		10.00
DL	0.01	0.01	0.01		0.01

4 - 06/13/2023: Corn, silage

4

Sample and source description: 4

Sample date: 10/04/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,200.00	2,800.00	13,000.00		4.90
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

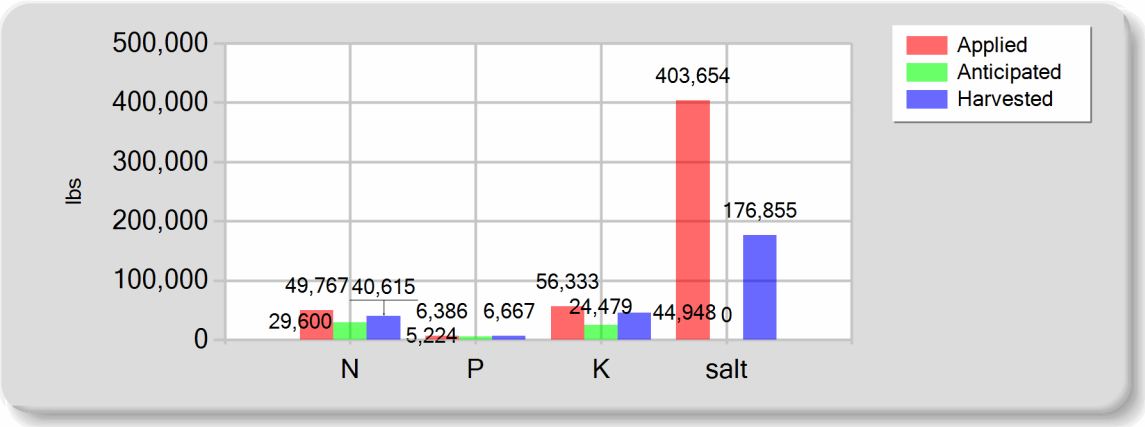
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

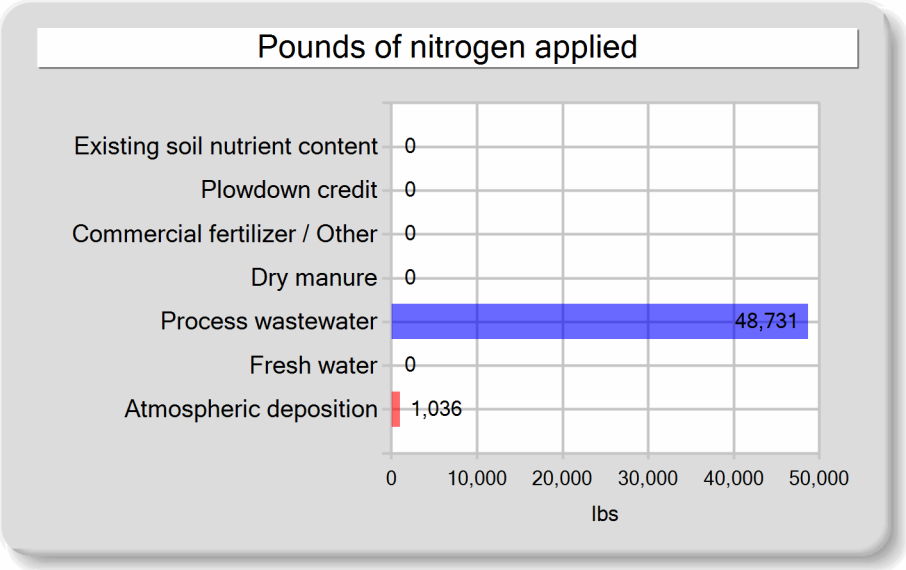
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

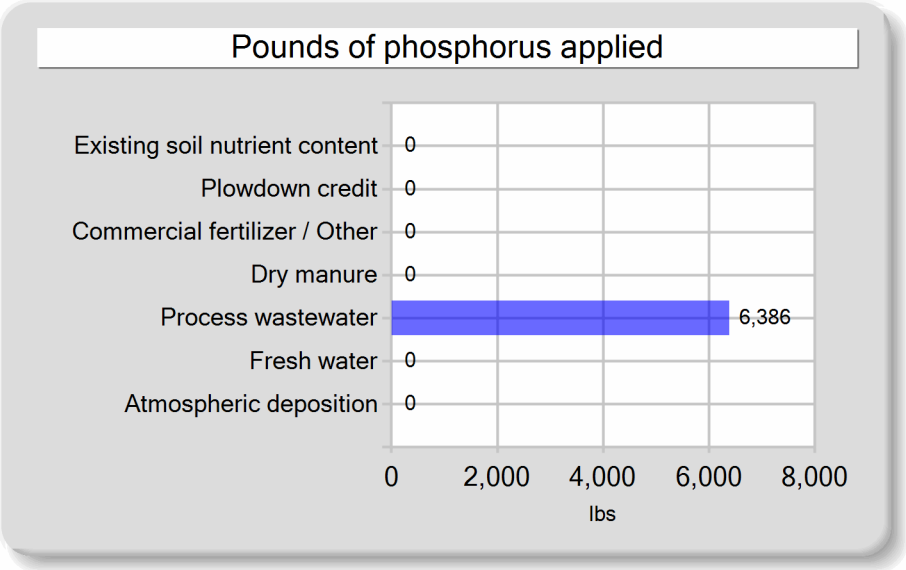
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	48,730.69	6,385.93	56,332.99	370,991.87
Fresh water	0.00	0.00	0.00	32,662.43
Atmospheric deposition	1,036.00	0.00	0.00	0.00
Total nutrients applied	49,766.69	6,385.93	56,332.99	403,654.30
Anticipated crop nutrient removal	29,600.00	5,224.40	24,479.20	0.00
Actual crop nutrient removal	40,615.18	6,667.16	44,947.87	176,855.04
Nutrient balance	9,151.51	-281.23	11,385.12	226,799.27
Applied to removed ratio	1.23	0.96	1.25	2.28

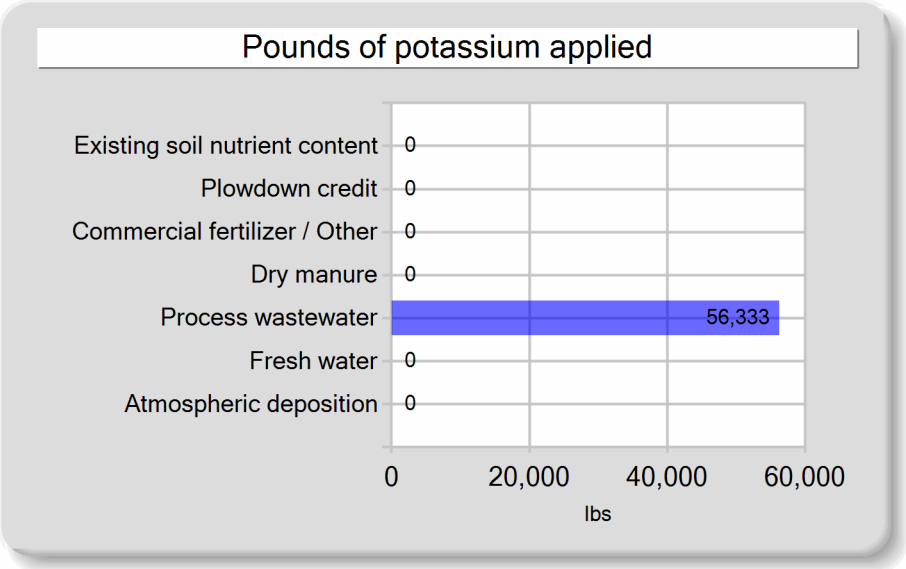
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

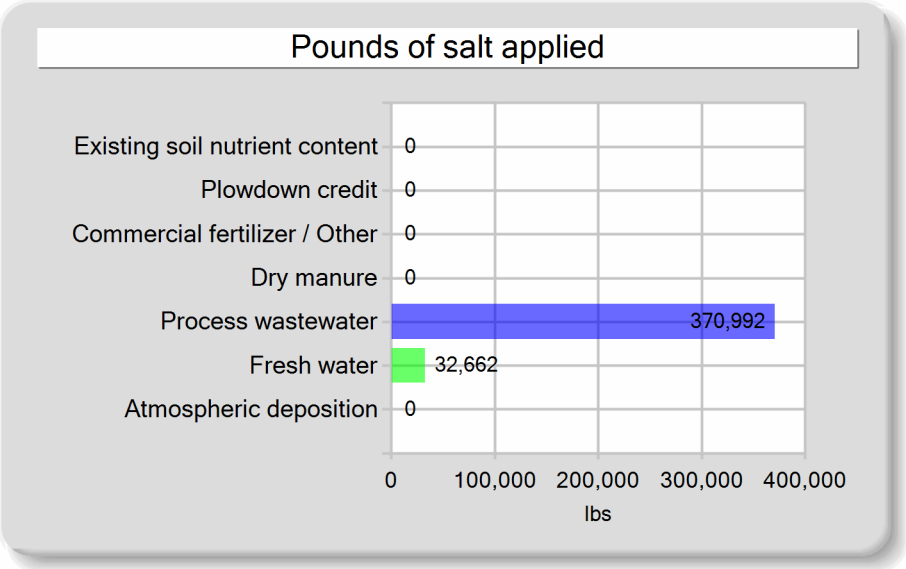


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

NO Animals were housed at this facility in 2023

*ALL Lagoon water that was applied on farm was Imported from a neighboring Dairy *

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Virgil Kronberg	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
6-24-24	
DATE	DATE

DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Kronberg Facility

Received: 08/03/2023 15:10
Reported: 08/07/2023 12:28

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23H0573-01	Dairy Well	Ag Water	T. Madruga		08/02/2023 16:00

Default Cooler Temperature on Receipt °C: -1.1
Containers Intact
COC/Labels Agree
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

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Ranch: Kronberg Facility

Received: 08/03/2023 15:10
Reported: 08/07/2023 12:28

Sample Results

Sample: Dairy Well
23H0573-01 (Water)

Sampled: 8/2/2023 16:00
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.14**	mmhos/cm	0.01	1		08/04/23 15:15	SM 2510 B		BEH0247
Electrical Conductivity umhos	**1140**	umhos/cm	10.0	1		08/04/23 15:15	SM 2510 B		BEH0247
Ammonia (as N)	*	mg/L	0.00	1		08/02/23 16:00	Field		BEH0312
Nitrate Nitrogen as NO3N	**23.0**	mg/L	0.1	1	10	08/04/23 20:32	EPA 300.0		BEH0235
pH	**7.7**	units	1.0	1		08/04/23 15:15	SM 4500-H+	H	BEH0247
Temperature	**25.0**	°C	0.0	1		08/04/23 15:15	SM 2510 B		BEH0247

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Submitted By: Rui & John Avila
Ranch: Kronberg Facility

Received: 08/03/2023 15:10
Reported: 08/07/2023 12:28

Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEH0235										
Blank (BEH0235-BLK1)					Prepared & Analyzed: 8/4/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEH0235-BLK2)					Prepared & Analyzed: 8/4/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
LCS (BEH0235-BS1)					Prepared & Analyzed: 8/4/2023					
Nitrate Nitrogen as NO3N	5.1		0.1	mg/L	5.000		102	90-110		
Duplicate (BEH0235-DUP1)		**Source: 23H0572-01**			Prepared & Analyzed: 8/4/2023					
Nitrate Nitrogen as NO3N	2.9		0.1	mg/L		2.9			0.988	10
Matrix Spike (BEH0235-MS1)		**Source: 23H0572-01**			Prepared & Analyzed: 8/4/2023					
Nitrate Nitrogen as NO3N	8.0		0.1	mg/L	5.000	2.9	101	90-110		
Reference (BEH0235-SRM1)					Prepared & Analyzed: 8/4/2023					
Nitrate Nitrogen as NO3N	10.2			mg/L	10.00		102	90-110		
Reference (BEH0235-SRM2)					Prepared: 8/4/2023 Analyzed: 8/5/2023					
Nitrate Nitrogen as NO3N	10.1			mg/L	10.00		101	90-110		

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Ranch: Kronberg Facility

Received: 08/03/2023 15:10
Reported: 08/07/2023 12:28

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEH0247										
Blank (BEH0247-BLK1)					Prepared & Analyzed: 8/4/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
pH	5.4		1.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Temperature	25.0		0.0	°C						
Blank (BEH0247-BLK2)					Prepared & Analyzed: 8/4/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
pH	7.5		1.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Temperature	25.0		0.0	°C						
Blank (BEH0247-BLK3)					Prepared & Analyzed: 8/4/2023					
pH	7.0		1.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	°C						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Duplicate (BEH0247-DUP1)		**Source: 23H0569-01**			Prepared & Analyzed: 8/4/2023					
Electrical Conductivity	1.52		0.01	mmhos/cm		1.54			1.70	10
pH	7.5		1.0	units		7.5			0.133	10
Electrical Conductivity umhos	1520		10.0	umhos/cm		1540			1.70	10
Duplicate (BEH0247-DUP2)		**Source: 23H0612-01**			Prepared & Analyzed: 8/4/2023					
pH	7.8		1.0	units		7.6			1.82	10
Electrical Conductivity	0.14		0.01	mmhos/cm		0.14			0.0713	10
Electrical Conductivity umhos	140		10.0	umhos/cm		140			0.0712	10
Reference (BEH0247-SRM1)					Prepared & Analyzed: 8/4/2023					
Electrical Conductivity	544			umhos/cm	538.0		101	90-110		
Reference (BEH0247-SRM2)					Prepared & Analyzed: 8/4/2023					
pH	7.9			units	7.790		101	.7163-101.28		
Reference (BEH0247-SRM3)					Prepared & Analyzed: 8/4/2023					
Electrical Conductivity	1020			umhos/cm	1000		102	90-110		
Electrical Conductivity umhos	1020			umhos/cm	1000		102	90-110		
Reference (BEH0247-SRM4)					Prepared & Analyzed: 8/4/2023					
Electrical Conductivity	1030			umhos/cm	1000		103	90-110		
Electrical Conductivity umhos	1030			umhos/cm	1000		103	90-110		
Reference (BEH0247-SRM5)					Prepared & Analyzed: 8/4/2023					
Electrical Conductivity	1040			umhos/cm	1000		104	90-110		

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Submitted By: Rui & John Avila
Ranch: Kronberg Facility

Received: 08/03/2023 15:10
Reported: 08/07/2023 12:28

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEH0247 (Continued)										
Reference (BEH0247-SRM5)					Prepared & Analyzed: 8/4/2023					
Electrical Conductivity umhos	1040			umhos/cm	1000		104	90-110		
Reference (BEH0247-SRM6)					Prepared & Analyzed: 8/4/2023					
pH	4.0			units	4.000		100	97.5-102.5		
Reference (BEH0247-SRM7)					Prepared & Analyzed: 8/4/2023					
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEH0247-SRM8)					Prepared & Analyzed: 8/4/2023					
pH	4.0			units	4.000		100	97.5-102.5		

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JG

08/03/23 15:10 23H0573

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 · 800 228-9896 · Fax 559 268-8174

No. Samples: 1 No. of Bottles: 1

Purchase Order No | Bill To: Acct # 19840 | Cons # 08

Results Need By

Water Type: [ ] Drinking Water [ ] Wastewater
[X] Ag Water [ ] Groundwater [ ] Monitoring Well

Name: R & J Avila Inc
Address: 8119 Ellington Avenue
City: Hilmar **State:** CA **Zip:** 95324
Telephone: (209) 669-7919 **Fax:** (209) 634-4604
Cell/Email: 209 678-3086 Rui / 209 678-1723 John frontdesk.randjavila@gmail.com
COPY TO:

REQUESTED BY: Rui & John Avila
PROJECT Kronberg Facility
ID CROP: N/A

[X] Copy of Chain [X] QA/QC Documents
Sampled By: T. Madruga

Other:

Analysis and Bottles Required: (Please indicate Analysis)

(X) **DWW1: EC, pH, NO_3-N (NH4-N Field Test)**
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DWW2: DWW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na, TDS
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DCW1: EC, NO_3-N, TDS + TKN & TN
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW1: EC, pH, NO_3-N, NH_4-N, TKN, TDS, TP, TK
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW2: DPW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) Other

	Description of Samples	Date Sampled	Time Sampled	Rec'd Temp °C	Field NH_4-N
1	Dairy Well	8/2/2023	1600	-1.1	
2					
3					
4					
5	IR Thermometer SN: 221314357				
6	Correction Factor: 0°C				
7	Calibration Due: 9/26/2023				
8	Location: Laboratory				
9					
10					

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	John Avila	R & J Avila, Inc.	8/2/2023 @ 1600	8/3/23 @ 0900
Second	JLF	Dellavalle Laboratory, Inc.	8/3/23 @ 0900	8-3-23 2:30
Third				
Fourth	[signature]	DLF	8/3/23 15:10	

I guarantee that as the client, or on behalf of client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a liquidated damage fee of 2% per month (annually 24%) or $5.00 per month whichever is greater.
If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (*cal*). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through ***cal*** under its Rules and Procedures. The parties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration,reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information: ***Shipping***
Sampling hrs $ In
Miles $ Out
Consulting

Amt Paid	Rec By	Check #	Date

Signature

Samples recieved in cooler with ice (coolant)
[ ] Yes [ ] No

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Shipping Information: Fed Ex ☐ UPS ☐ GSO ☐ Picked-Up ☑ Walk In ☐ DLI Sampler ☐
Ontrac ☐ PMS ☐ USPS ☐ Other ☐ _____

Shipping Container: Ice Chest ☑ None ☐ Box ☐

Refrigerant: Wet Ice: ☑ Blue Ice: ☐ None: ☐

Container(s) Received	1	2	3	4	5	6	7	8	9	10
Plastics:										
100 mL sterile plastic $Na_2S_2O_3$ (Green)										
250 mL unpreserved (White)										
250 mL HNO_3 (Red)										
250 mL H_2SO_4 (Yellow)										
500 mL unpreserved (White)										
500 mL HNO_3 (Red)										
500 mL H_2SO_4 (Yellow)										
1 L unpreserved (White)	1									
1 L unpreserved (BOD) (Purple)										
1 L HNO_3 (Red)										
1 L H_2SO_4 (Yellow)										
VOA Vials in sets of 3:										
40 mL VOA, HCl (Blue)										
40 mL VOA, $Na_2S_2O_3$ (Green)										
40 mL VOA, unpreserved (White)										
Glass:										
250 mL AG unpreserved (White)										
250 mL AG H_2SO_4 (Yellow)										
250 mL AG $Na_2S_2O_3$ (Green)										
250 mL AG $Na_2S_2O_3$ + MCAA										
500 mL glass unpreserved (White)										
500 mL AG HCl (Blue)										
1 L AG unpreserved (White)										
1 L AG H_2SO_4 (Yellow)										
1 L AG $Na_2S_2O_3$ (Green)										
1 L AG HCl (Blue)										
Special:										
Cr^{6+} w/Borate/HCO3/CO3										
Cyanide - 500 mL NaOH										
Asbestos - 2 - 1L P wrapped in foil										
500 mL or 1 L **AP** $Na_2S_2O_3$ (Green)										
1 L **AG** or **P** NaOH + ZnAc (Sulfide)										
Chlorite/Bromate 250 mL AG with EDA										
TTHM										
HAA5										
40 mL VOA, $Na_2S_2O_3$ + MCAA										